

ALMOST EQUIVALENT: MEASURING WHEN DOWNSCALED IMAGES CARRY THE FLOW-MATCHING TRAINING SIGNAL

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ABSTRACT

Scale-wise diffusion schemes rest on a spectral argument: flow-matching noising progressively masks high spatial frequencies, so above some noise level a down-scaled latent carries *almost* the full surviving signal. We ask whether this licenses a training-side substitution — can the gradient of a low-rank adapter be computed from a downscaled latent at high noise without changing what is learned? We give a first-order account of what the substitution can cost: the gradient gap decomposes into a noise-dependent *input term*, read through the gradient norm, plus a σ -independent *floor* carried by the network’s input–output Jacobian — and we embed the spectral argument, in its strongest tolerance-parameterized form (Xiao et al., 2026), as the null model of the input term: a one-parameter family of per-route boundaries that depends on the downscale *ratio* alone, is compressed to a ~ 0.13 spread in σ on our measured latent spectrum, and admits no floor. Testing this takes metrology, not just measurement: naive finite-draw gap estimators *manufacture* exactly the token-count-ordered floors at issue, so we build a debiased estimator (per-arm self-floors with an attenuation correction, plus draw-count extrapolation) and state every claim at the instrument’s measured resolution. The debiased measurements on a DiT-based flow-matching model reject the null where the accounts diverge: the mildest route’s crossover sits at $\sigma \approx 0.5$, $\sim 3.5\times$ the equal-power prediction; a $2\times$ downscale is certifiably unsafe at *every* noise level, retaining a debiased gap of ≈ 0.30 on pure-noise input; and the surviving endpoint gap equals its image-free control at every route — a floor carried by the compute graph, governed by the *absolute token count* of the downscaled grid, concentrated in early blocks, and separable into a rotary-position-embedding share and a graph residue. Debiasing also cuts both ways: about half of the intermediate route’s raw floor was estimator bias; what remains ($+0.056$, CI $[0.043, 0.071]$) is real but leaves no certifiable safe window at our resolution. The practical output is a measured (route, σ) safety map with explicit detectability thresholds, and an opt-in σ -conditional trainer that substitutes the downscaled latent only on measured-safe steps, for a projected wall-clock ceiling of $\sim 14\%$ at fixed step count on our corpus (end-to-end A/B pending).

1 INTRODUCTION

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. A family of recent methods therefore runs part of the computation at reduced resolution, justified by a spectral observation about the forward noising process: under flow-matching noising $z_\sigma = (1 - \sigma)x + \sigma\epsilon$, the per-frequency signal-to-noise ratio falls with σ , and high spatial frequencies drop below the noise floor first. Above a crossover noise level, a downscaled latent is claimed to carry *almost* the full surviving signal — the premise behind scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), pyramidal flow matching (Jin et al., 2024), spectrally-guided noise schedules (Esteves & Makadia, 2026), and progressive-resolution training schedules more broadly (Karras et al., 2018; Chen et al., 2024; Hooigeboom et al., 2023).

The evidence for these methods is collected where they operate: at the level of the generated output. Reduced-resolution computation, suitably scheduled, can drive the *quality-score* difference against

full-resolution computation nearly to zero. Our central finding is that this success statement does not transfer to training: a small quality-score difference does not imply a small difference in what a training step *learns*. On our latents the spectral account, computed from the measured radially-averaged power spectrum, predicts that a downsampled latent is a safe substitute from $\sigma \approx 0.14$ (and by $\sigma \approx 0.6$ at any published tolerance); the measured *training-signal* boundary for even the mildest downscale is $\sigma \approx 0.5$, and no harsher route certifies at any noise level — the 2 $\times$ downscale retains a debiased gap of ≈ 0.30 even on pure-noise input (§5.1, Fig. 1). Existing low-resolution training can make the quality gap small; it does not make the learning gap small.

Our goal is therefore to derive the regime — if one exists — in which training on downsampled data changes the training signal by almost nothing. This is a narrower, training-side question than the inference-side ones above: at high noise, can the *gradient* of an adapter be computed on a downsampled latent as a drop-in substitute for the native one — same model, same objective, same σ distribution, only the spatial grid changed on some steps? The payoff would be large: on our corpus roughly half of all training draws sit at $\sigma > 0.5$.

Rather than measuring first and rationalizing after, we first state what the substitution can cost, then build an instrument whose resolution is itself measured, and then let the measurements discriminate. The contributions:

1. **Gap metrology: a debiased estimator and a detectability threshold** (§3). The natural gap estimator — redraw floor minus a single demoted-arm cosine — is biased by finite-draw variance, and the bias *grows as the demoted grid shrinks*: it manufactures exactly the token-count-ordered floor signature under test. An uncorrected native self-cosine reads 0.85 at the noise endpoint; corrected, it extrapolates to 1.005 (CI [0.994, 1.016]). We introduce per-arm self-floors with a split-half attenuation correction and a draw-count extrapolation $\text{gap}(D) = \text{gap}_\infty + c/D$, and derive the instrument’s minimum certifiable margin $\varepsilon^*(N, D, \text{floor})$; “safe” is a one-sided non-inferiority statement at that resolution, never an absolute claim.
2. **A two-term account of the demotion gap, stated before it is fit** (§4). Writing the adapter gradient as $g = J^T r$, demotion perturbs exactly two things — the information the input still carries, and the compute graph itself — giving the first-order form $\text{gap}_e(\sigma) \approx A_e m(\sigma)/G(\sigma)^p + \text{Floor}_e$: a noise-dependent input branch read through the U-shaped gradient norm, plus a σ -independent floor. The spectral argument, in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026), is embedded as the null model of the input branch — a null that on our latent spectrum depends on the downscale *ratio* alone, spans only ~ 0.13 in σ across all tolerances, and admits no floor.
3. **Debiased measurements that reject the null where the accounts diverge** (§5, §6). The mildest route’s crossover sits at $\sigma \approx 0.5$ — no tolerance reproduces the map — and a 2 $\times$ downscale is certifiably unsafe at every noise level, retaining a debiased gap of ≈ 0.30 on pure-noise input (12/12 probe images above the pre-registered 0.15 bar). The endpoint gap is statistically identical to its image-free control at every route: the floor is pure graph term — σ -independent, governed by absolute token count rather than ratio, concentrated in early blocks, and split — via a positional-interpolation intervention — into a rotary coordinate-system share and a graph residue; the prediction-residual mismatch is *route-uniform*, placing all route identity in the Jacobian. Debiasing also revises us: half the intermediate route’s raw floor was estimator bias, and its “never safe” verdict softens to “no certifiable window at instrument resolution.”
4. **A measured safety map and a practical trainer** (§7). Safety is a per-route boundary $\sigma^*(\text{route})$ at a stated resolution, not a universal ratio rule: {1024 $\rightarrow$ 896 at $\sigma > 0.5$; 1280 $\rightarrow$ 1024 at $\sigma \gtrsim 0.75$ }, while 1024 $\rightarrow$ 768 has no certifiable window and every $\rightarrow$ 512 route is certifiably unsafe everywhere. An opt-in σ -conditional trainer draws the noise level first and substitutes the downsampled latent only on measured-safe steps, for a projected wall-clock ceiling of $\sim 14\%$ at a fixed step budget (end-to-end A/B pending).

The presentation is theory-first; the practice was observational-first: verdict criteria were frozen before each probe ran, the two-term account was pre-registered with an outcome matrix (including a kill switch) before the discriminating runs, and the debiasing campaign’s decision rules — including the rule under which our own headline floors would have been retracted as variance artifacts — were

pre-registered before it ran. Each mechanistic claim is grounded in a designated falsification test (several of which fired and killed proposed routes; we report those too).

2 RELATED WORK

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogetboom et al., 2023) train at increasing resolution as a schedule. Pyramidal flow matching (Jin et al., 2024) and scale-wise distillation (SwD) (Starodubcev et al., 2025) tie resolution to the noise level itself, with SwD giving the spectral analysis we take as the hypothesis under test: their Fig. 1-style argument shows the noised downsampled latent is close in distribution to a downsampled noised latent above a crossover σ . Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives that argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model — a criterion on the *Bayes-optimal predictor’s* per-band output, not merely on input SNR — from which an inference-time resolution schedule follows; we adopt their Proposition 1 as the null model of our input branch (§4.3). SPD also fine-tunes (with LoRA) against *stage-specific modified velocity targets* along the progressive trajectory — changing the objective per resolution stage rather than substituting the latent under the native objective, which is our question; that they observe a training–inference gap motivating this fine-tuning is independently consistent with our finding that resolution changes are not free at the training level. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same per-frequency reasoning to schedule design, deriving bounds on the useful minimum and maximum noise levels from the image spectrum — the spectral account operating as the field’s working model of what noise masks. We are explicit about scope: neither SPD nor SwD *claims* naive gradient equivalence under latent substitution — SwD validates its pipeline at the output level, and SPD fine-tunes precisely because resolution changes are not free. What we test is the tempting extension of their shared premise — that spectral sufficiency of the input licenses a training-side swap — not their methods. Our contribution is orthogonal to all of these: we measure whether input-level closeness transfers to the *training gradient*, holding model and objective fixed — and find it does not transfer at any tolerance.

Flow matching and DiT. We work in the flow-matching / rectified-flow formulation (Lipman et al., 2023; Liu et al., 2023; Esser et al., 2024) on a Diffusion Transformer (Peebles & Xie, 2023) with rotary position embeddings (Su et al., 2024). Adapters are LoRA (Hu et al., 2022).

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* — matching the downsampled grid’s relative phase geometry to native — to decompose the resolution-sensitivity floor, and adopt the σ -gated band schedule of Zhao et al. (2026) (functional form only; their spectral scale laws mispredict our measured boundaries, consistent with our main finding) as a training-time refinement.

Gradient-agreement methodology. Reading training decisions off gradient cosines against a noise-redraw floor follows the general logic of gradient-noise-scale analyses (McCandlish et al., 2018); our estimator operates per σ -bin with means over images, the regime we found reliable (split-half reliability 0.73–0.88), rather than per-image rankings, which were pure noise in our earlier attempts.

3 THE DEMOTION GAP AND WHAT IT CAN RESOLVE

Every claim in this paper is stated in one metric — the demotion gap — and at a stated resolution. This section defines the metric and the instrument, exhibits the estimator-variance confound that a naive read of the metric suffers (it manufactures exactly the floor signature under test), constructs the debiased estimator that removes it, and derives the detectability threshold ε^* against which “safe” is defined.

Objective and noising. With clean latent x , noise $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma \in (0, 1]$, the noised input is $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ and the flow-matching target is $v = \epsilon - x$; the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$ with gradients taken with respect to adapter parameters only.

Demotion. A *demotion* $e_{\text{nat}} \rightarrow e$ replaces the native latent with one produced by pixel-space downscaling of the source image to the tier- e bucket followed by VAE re-encoding — the ordering SwD validated (“strategy B”); we never downsample in latent space. The caption, the noise draw structure, and the σ under test are held fixed; only the spatial grid changes.

The verdict quantity. For a set of images and a σ -bin, adapter gradients are accumulated per arm and compared by cosine; the *demotion gap* is

$$\text{gap}_e(\sigma) = \cos_{\text{floor}}(\sigma) - \cos(g_{\text{nat}}(\sigma), g_e(\sigma)), \quad (1)$$

where $\cos_{\text{floor}}$ is the same-bin cosine between two independent native draw sets — the “how much do gradients differ anyway” null. The re-encoding control below prices the VAE round trip that demotion necessarily pays; §4 asks what structure Eq. 1 can have.

3.1 INSTRUMENT

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier- e image occupies roughly $(e/16)^2$ latent-patch tokens (e.g. $\sim 4,100$ tokens at 1024, $\sim 3,000$ at 896, $\sim 2,160$ at 768, $\sim 1,000$ at 512). The probe adapter is a plain LoRA checkpoint trained at native tiers; we return to this operating-point dependence in §8.

The gradient probe. For each image and each of B uniform σ -bins we accumulate adapter gradients over D stratified noise draws per arm (verdict runs: 8×8 unless noted; $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients (Eq. 1):

- **floor:** two independent draw sets at native resolution — the redraw null;
- **reenc:** decode–re-encode at native resolution — isolates the VAE round-trip cost that demotion necessarily pays;
- **demote arms:** one per edge e under test;
- **self-floors** (debiased runs): a second independent draw set for *every* arm, giving each arm the same two-estimate structure the native floor has (§3.2).

The gap is in cosine units (a mismatch *fraction*, scale-free; bin-mean SEM ~ 0.02 at $N=40$). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images — a discipline forced on us by an earlier failure in which per-image rankings from the same gradients had split-half reliability indistinguishable from zero. Uniform σ -bins make the training-time σ -density irrelevant to the map; density-weighting the bins reproduces our earlier pooled (marginalized) measurements as a consistency check. Estimator details and the endpoint / x-zero probe modes are specified in Appendix A.

3.2 FINITE DRAWS MANUFACTURE FLOORS

Eq. 1 as naively estimated is asymmetric: the floor is a cosine between *two* finite-draw estimates of the native gradient, while each demote arm contributes *one* estimate compared against the native mean. Finite-draw noise attenuates every such cosine below its population value, and the attenuation is arm-dependent: per-draw gradient variance grows as the token count falls (the MSE loss averages over fewer elements), so a demoted arm’s single estimate is noisier than the native estimates it is compared to. An iso-direction null — demotion changing *nothing* but the variance — therefore still produces a positive gap that grows as the target grid shrinks: the same signature as “absolute token count sets the floor.” A back-of-envelope with the raw endpoint floor and noise $\propto 1/\text{tokens}$ predicts spurious

endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512 — not the whole of the raw measurements ($-0.01/0.13/0.33$), but plausibly $\sim 40\%$ of the intermediate floors. A retroactive scan of earlier runs confirms the confound is live: the 1024 $\rightarrow$ 896 endpoint gap decays as $+0.100/+0.035/-0.016$ at $D = 4/8/16$ — a clean c/D decay with fitted $c \approx 0.5$. No conclusion in this paper survives on the naive estimator alone.

The native floor itself makes the point sharpest: at $D=8-16$ the raw endpoint self-cosine reads 0.79–0.85, but a nested draw sweep ($D = 4 \dots 64$) extrapolates it to 1.005 (bootstrap 95% CI [0.994, 1.016]). The population-level per-bin native gradient is a *fixed direction*; the entire redraw floor is finite-draw noise, not intrinsic gradient stochasticity.

The debiased estimator. Two corrections, used together:

1. **Self-floors with attenuation correction.** Every arm runs a second independent draw set g'_d , giving a per-arm self-cosine $\cos_{\text{self}} = \cos(g_d, g'_d)$. The debiased cosine is the classical split-half disattenuation (Spearman, 1904),

$$\hat{c} = \frac{\cos(\bar{g}_{\text{nat}}, \bar{g}_d)}{\sqrt{\cos_{\text{floor}} \cdot \cos_{\text{self},d}}}, \quad \widehat{\text{gap}} = 1 - \hat{c}, \quad (2)$$

which cancels the finite-draw attenuation of numerator and denominator to first order. Raw gaps are reported alongside for continuity.

2. **Draw-count extrapolation.** Endpoint-mode runs sweep nested draw prefixes $D \in \{4, 8, 16, 32, 64\}$ (the $D=64$ pass contains the smaller draw sets — one set of forwards) and fit $\text{gap}(D) = \text{gap}_{\infty} + c/D$ per route, with a bootstrap CI over images. The two corrections cross-validate: debiased fits are D -flat ($|c| \leq 0.05$ for the 512 route, against $c \approx +0.29$ raw) — the attenuation correction removes exactly the $1/D$ component the sweep isolates.

One caveat delimits where Eq. 2 is readable. At $D=8$ per bin, the *unpaired* debiased estimator is unstable in bins where the floors themselves are small (mid- σ ; the denominator is noisy): there the readable object is the *paired per-image difference* against the re-encoding control, $\Delta_i = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$, with non-finite and $|\Delta| > 1.5$ values trimmed — pairing cancels the shared per-image, per-bin draw structure. All verdict maps below report this paired object. Finite-draw cosines are also compiler-kernel-path sensitive, so all pairings live within one run (Appendix A).

3.3 ε^* : WHAT THE INSTRUMENT CAN CERTIFY

A safety claim needs a stated resolution. For a route whose true gap is zero, the paired difference Δ fluctuates around zero with standard error $\text{SE}(N, D, \sigma\text{-bin})$, so the one-sided 95% upper bound $\bar{\Delta} + 1.645 \text{ SE}$ concentrates near 1.645 SE: no margin smaller than

$$\varepsilon^*(N, D, \text{bin}) = 1.645 \text{ SE}(N, D, \text{bin}) \quad (3)$$

is certifiable even for a truly safe route. We therefore *define* “safe at margin ε ” as the non-inferiority statement

$$\text{UB}_{95}(\bar{\Delta}) < \varepsilon, \quad \varepsilon \geq \varepsilon^*, \quad (4)$$

replacing the ad-hoc ± 0.04 re-encoding band of our earlier drafts. Measured values: at the verdict grid’s operating point ($N=40$, $D=8/\text{bin}$) the bin-level ε^* is 0.03–0.08; endpoint-mode draw sweeps (D up to 64) tighten it to ≈ 0.01 –0.02. Two consequences are worth stating before any measurement. First, a “safe” verdict is always a statistical non-inferiority claim at instrument resolution — never a hard bound; a true gap below ε^* is undetectable by construction. Second, ε^* is the honest currency for negative claims too: “no certifiable window” means the route’s upper bounds never cross below ε at the current (N, D) , which a larger campaign could still change — whereas “certifiably unsafe” (lower bound above ε) cannot.

4 WHAT DEMOTION CAN COST: A TWO-TERM ACCOUNT

4.1 WHY INPUT-LEVEL SPECTRAL SUFFICIENCY CANNOT BOUND THE GRADIENT

The spectral argument is a statement about the *input*. Two observations, neither involving the spectrum of z_{σ} , show it cannot bound the training gradient:

(a) Expected gradients are never noise-masked. “Noise power exceeds signal power in band f ” is a statement about a single sample of z_{σ} . The quantity training consumes is $\mathbb{E}_{\epsilon}[\nabla_{\theta} \mathcal{L}]$ — an expectation that averages the noise out and retains attenuated signal at *any* SNR. Input-level sufficiency licenses inference-time substitution; it never licenses training-dynamics equivalence.

(b) The target carries the clean image at unit amplitude at every σ . The flow-matching target is $v = \epsilon - x$. The *input* is progressively noise-masked as $\sigma \rightarrow 1$, but x sits in the *target* at coefficient one even at $\sigma = 0.94$. Writing the adapter gradient as $g = J^T r$ (input–output Jacobian times prediction residual), the two grids can disagree through J itself — attention softmax over a different token count N , rotary phase density, sequence-length-dependent normalization — regardless of how little image signal the input retains.

Together these say a demoted input can disagree with the native gradient through two distinct channels: what the input still *carries* (information), and what the network *does* with a different grid (computation). The rest of this section turns that observation into a measurable form.

4.2 THE FIRST-ORDER FORM

Demotion changes exactly two things about the map from (image, noise, caption) to adapter gradient:

1. **the input branch:** spatial-frequency bands above the coarse grid’s Nyquist limit are absent from input *and* target (and the re-encode perturbs what survives) — a perturbation $\Delta r(\sigma)$ of the prediction residual whose size is governed by posterior physics and shrinks as the model’s reliance on input detail shrinks with σ ;
2. **the graph:** token count, rotary phase density, sequence-length-dependent normalization — an operator perturbation ΔJ that is σ -*independent*, because the grid change does not reference σ .

To first order the gradient perturbation is $\delta g \approx J^T \Delta r(\sigma) + \Delta J^T r$, with the cross term second-order. One qualification is worth stating precisely: J_{nat} and J_e act on different grids, so the two terms are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live. This is exactly what the designated probes of §5.2 isolate: the $\sigma=1$ endpoint makes the input branch vanish in distribution (the input is pure ϵ on every grid), and the x-zero probe removes the image from input and target entirely — each isolating the graph term by construction.

Cosine geometry then fixes how a mismatch is read. For a perturbation δg with component $\delta g_{\perp}$ orthogonal to g ,

$$1 - \cos(g, g + \delta g) \approx \frac{1}{2} \|\delta g_{\perp}\|^2 / \|g\|^2 \quad (5)$$

in the small-mismatch limit, degrading toward $\|\delta g_{\perp}\|/\|g\|$ when the mismatch is comparable to the signal. Any absolute mismatch is therefore read through the total gradient scale $G(\sigma) = \|g\|$, with an exponent pinned to $p \in [1, 2]$ by the geometry alone — 2 in the small-orthogonal-mismatch limit. Under three assumptions — **(A1)** the two perturbations are uncorrelated in parameter space (no interaction term), **(A2)** the graph term’s *relative* footprint $\|\Delta J^T r\|/\|g\|$ is σ -stationary (its numerator and denominator share the residual’s scale), and **(A3)** the input branch’s gain through J^T is σ -independent, so it factorizes into a route amplitude times a σ -shape — the gap takes the two-term form

$$\text{gap}_e(\sigma) \approx \underbrace{A_e \cdot m(\sigma) / G(\sigma)^p}_{\text{input branch } S_1} + \underbrace{\text{Floor}_e}_{\text{graph term}}, \quad p \in [1, 2], \quad (6)$$

where $m(\sigma) = \|\Delta \bar{r}(\sigma)\|$ is the mismatch of the model’s mean prediction residual across grids (directly measurable, §5.3), A_e is a σ -independent route amplitude absorbing units and the Jacobian gain, and Floor_e is the σ -independent term carried by the network function itself.

The ledger, explicitly: the *form* — additive two-term structure, the G^p denominator, $p \in [1, 2]$, the σ -flatness of the floor — is derived from the perturbation expansion plus cosine geometry, conditional on A1–A3; the assumptions are tested by designated probes (§5.2); the coefficients A_e , $m(\sigma)$, $G(\sigma)$, Floor_e are measured, not derived. The account was pre-registered with an outcome matrix (including a kill switch: all endpoint gaps ≈ 0 would have falsified the graph term) before the discriminating runs.

4.3 THE SPECTRAL ACCOUNT AS A NULL MODEL OF THE INPUT BRANCH

The spectral argument, made precise, is a model of the input branch alone — and its strongest available form is SPD’s tolerance-parameterized crossover (Xiao et al., 2026). Under the modeling assumption $x_0^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ per frequency ω (with P the clean-latent power spectrum), their Proposition 1 states that for all $\sigma \geq t_\omega$ the *Bayes-optimal* per-band velocity prediction is within tolerance δ of pure noise, $\mathbb{E}[|v^{*(\omega)} - \epsilon^{(\omega)}|^2] \leq \delta$, at

$$t_\omega = \frac{1}{1 + \sqrt{\delta / (P_\omega (1 + P_\omega - \delta))}}, \quad (7)$$

and their Proposition 2 places the safe-substitution boundary for a coarse grid at the t_ω of its Nyquist band. This is a prediction-level criterion, strictly stronger than raw input SNR; the often-quoted equal-power crossover $\sigma_{\text{eq}}(f) = \sqrt{P(f)/(1 + \sqrt{P(f)})}$ is its $\delta = (1 + P_\omega)/2$ slice.

Taken as the null model of S_1 , Eq. 7 has three structural properties, each checkable before any gradient is measured:

(N1) One-parameter family. All route boundaries move together under the single tolerance δ ; the family is totally ordered.

(N2) Ratio-only. In normalized frequency the demoted grid’s Nyquist sits at $f_{\text{cut}} = \frac{1}{2} e/e_{\text{nat}}$ — a function of the downscale *ratio* alone. The null therefore cannot distinguish routes of equal ratio at different absolute sizes, and assigns nearly identical boundaries to nearly equal ratios regardless of grid size. It is structurally blind to any absolute-size effect.

(N3) No floor. For every route and every δ , the null’s gap vanishes above a finite boundary: every route becomes safe at some σ .

And under the same diagonal-Gaussian assumption the null extends to the residual level: the posterior mean is a per-band Wiener shrinkage — under a Gaussian prior and Gaussian noise the minimum-mean-square-error estimator is linear with per-band gain $P_\omega/(P_\omega + n_\omega)$, the classical Wiener filter (Wiener, 1949) — so with no cross-frequency coupling the cross-grid mean-residual mismatch is confined to the destroyed band and **(N4)** ordered across routes by destroyed-band energy.

The family is instantiated on our measured latent spectrum, and confronted with the measurements in gap units, in §6 (Table 5). Two properties of that instantiation are worth previewing because they are fixed before any gradient is measured: the latents are high-frequency-quiet, so the family is *compressed* — the spread between the mildest and harshest route never exceeds ~ 0.13 in σ at any tolerance — and the two routes our measurements will separate maximally (1024→896, ratio 0.875, and 896→768, ratio 0.857) receive boundaries within 0.01 of each other at *every* δ .

4.4 THE GRAPH TERM: A POSITIONAL SHARE AND A GRAPH RESIDUE

The operator perturbation ΔJ has identifiable parts. Changing the grid changes (i) the rotary coordinate system — every band of the position embedding accumulates phase at a different density across the image — and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the sheer capacity of the coarse graph to approximate the fine graph’s computation. Splitting $\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$ yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention — evaluating the demoted grid at fractional rotary coordinates that reproduce the native relative phase geometry — should remove RoPE_e wherever the input branch is inactive ($\sigma \rightarrow 1$), and a band-counting sketch says the residue should grow with absolute grid change (the number of rotary bands whose per-extent rotation count crosses decorrelation grows with the extent), attaching the absolute-size governor to Resid_e . We flag the sketch’s epistemic status plainly: like YaRN’s own band thresholds (Peng et al., 2024), its constants are calibrated, not derived; we claim no closed form for Floor_e .

4.5 WHERE THE ACCOUNTS DIVERGE

The two-term account (Eq. 6) and the spectral null (Eq. 7) agree that gaps shrink with σ and order by severity. They diverge on five measurable axes, which the experiments of §5 score:

Table 1: **Raw estimator** (historical record): demotion gap by σ -bin, 1024-tier natives ($N=40$, bin-mean SEM ~ 0.02 , re-encoding control $|\text{gap}_{\text{reenc}}| \leq 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. The equal-power slice of the spectral family predicts collapse above the second bin for every route; no other slice fits either (Table 5). Raw values understate mid- σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); Table 2 corrects both.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap ₈₉₆ (ratio 0.875)	.110	.162	.148	.137	.048	.030	.030	.053
gap ₇₆₈ (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap ₅₁₂ (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
cos _{floor}	.84	.65	.51	.65	.70	.77	.80	.83
$\ g\ $ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

1. **Boundary pattern:** the null predicts some row of Table 5; the account permits {finite boundary, never, never} if floors differ (§5.1).
2. **High- σ persistence:** the null predicts every gap vanishes as $\sigma \rightarrow 1$; the account predicts persistence equal to Floor_e, surviving even pure-noise input (§5.2).
3. **Governors:** the null is ratio-only (N2); the account permits ratio to govern A_e while absolute size governs Floor_e (§5.4).
4. **Residual ordering:** the null predicts route-ordered residual mismatch (N4); if route identity instead lives in J , the residual curves are route-uniform under route-ordered gaps (§5.3).
5. **Curve shape:** the null ties the gap’s shape to per-band SNR; the account ties it to $m(\sigma)/G(\sigma)^p$ — an interior peak where the U-shaped G is smallest, and a low- σ dip that renormalizing by G removes (§5.3).

5 MEASUREMENTS

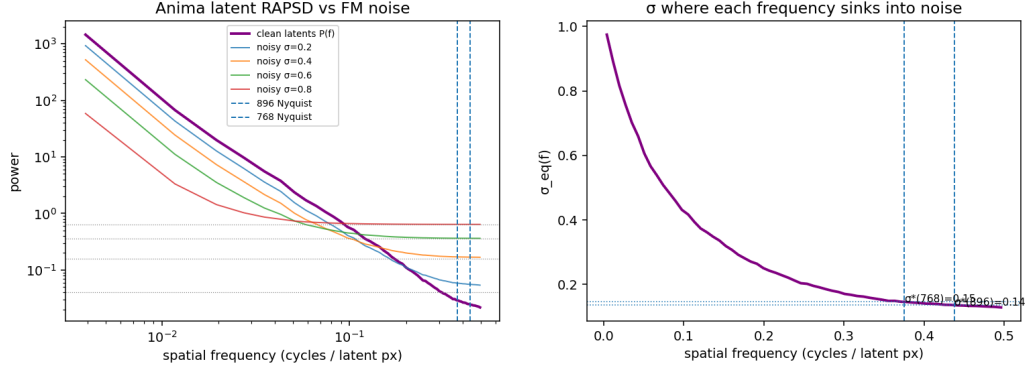
Reporting convention: headline floors and the verdict map are stated in *debiased* units — draw-limit values $\widehat{\text{gap}}_\infty$ with bootstrap 95% CIs from the endpoint sweeps, and the paired per-image object (§3.2) for per-bin maps. Raw single-estimate values are retained where they are the historical record (the original verdict-grid curves, and secondary probes whose conclusions rest on *paired* same-grid contrasts that cancel the finite-draw bias to first order); every raw number is marked as such.

5.1 THE GAP CURVES VERSUS THE SPECTRAL FAMILY

The spectrum. Anima’s VAE latents are high-frequency-quiet ($P(f) < 1$ above $f \approx 0.16$ cycles/latent-pixel; latent variance 0.42), giving the equal-power crossovers $\sigma^*(896) = 0.136$, $\sigma^*(768) = 0.146$, $\sigma^*(512) \approx 0.20$, with tight per-image spread (p10–p90 ≈ 0.11 –0.16): the spectral crossover is image-generic, and at that tolerance demotion would be safe over almost the entire σ axis (Fig. 1, top row). The full tolerance family these cuts admit is Table 5.

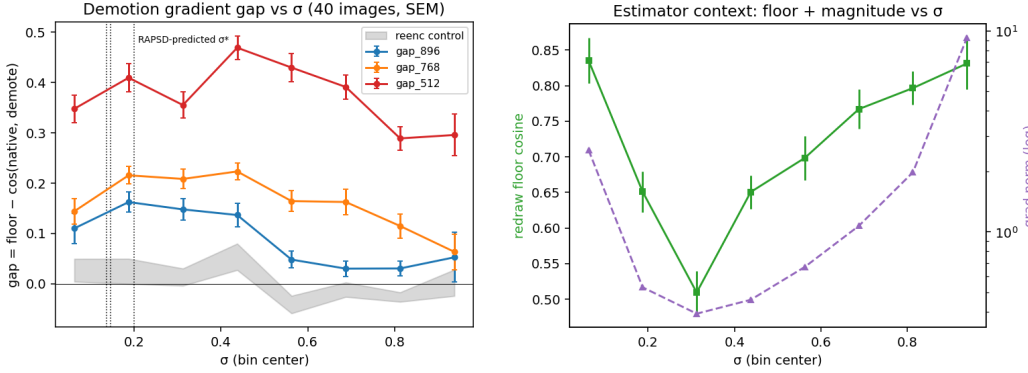
The measurement. Table 1 gives the original (raw-estimator) gap curves for 1024-tier natives demoted to 896/768/512; Table 2 gives the debiased re-measurement of the same grid — the verdict object.

Three reads, against criteria frozen before the runs: (i) σ -dependence is real and in the predicted direction (raw Spearman $\sigma \rightarrow \text{gap}$: $-0.69/-0.57/-0.36$), and tier ordering holds at every bin — the *qualitative* spectral intuition survives; (ii) the crossover for the mildest route is $\sigma \approx 0.5$ in raw *and* debiased units (debiased 896: clearly gapped through $\sigma \leq 0.31$, marginal at 0.44, means within ± 0.03 of zero from 0.56 on) — $3.5\times$ the equal-power prediction, and no tolerance rescues the family: the δ that matches this boundary ($\delta = 0.025$, Table 5) predicts 768 safe at 0.52 and 512 safe at 0.62, while SPD’s own published $\delta = 0.01$ predicts all three routes safe by 0.72; (iii) the $2\times$ downscale route is *certifiably unsafe at every noise level*: its paired debiased lower bound stays above ε^* in every bin (means $+0.08$ to $+0.60$), including $+0.17 \pm 0.06$ at $\sigma = 0.94$, where the input is $\sim 97\%$ noise by



(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts.

(b) The equal-power slice $\sigma_{eq}(f)$ of the spectral family. Predicted σ^* : 0.136 / 0.146 / ~ 0.20 for demotion to 896 / 768 / 512.



(c) Measured demotion gap per σ -bin (1024-tier natives, $N=40$; raw estimator — debiased map in Table 2). The 896 route floors at $\sigma \approx 0.5$; 512 never floors.

(d) Estimator context: the redraw-floor cosine and the U-shaped gradient norm $G(\sigma)$ through which every gap is read (§5.3).

Figure 1: The spectral account (top row: computed from the measured RAPSD, predicting safety from $\sigma \approx 0.14$) versus the gradient measurement (bottom row: safe only from $\sigma \approx 0.5$, and only on the mildest route). No tolerance reproduces the measured pattern (Table 5): the equal-power crossover is off by $\sim 3.5\times$ and the $2\times$ downscale never becomes safe.

Table 2: **Debiased verdict map**: paired per-image debiased gap against the re-encoding control, $\bar{\Delta}$ (SEM), 1024-tier natives ($N=40$, $D=8/\text{bin}$ plus a $\sigma=1$ endpoint bin at the same draws; non-finite and $|\Delta| > 1.5$ per-image values trimmed, retaining 35–40 images per cell). Bin-level ε^* at this (N, D) is 0.03–0.08 (Eq. 3). Debiasing *raises* the mid- σ gaps the raw read attenuated and *lowers* the endpoint floors the raw read inflated; the 896 crossover at $\sigma \approx 0.5$ and the 512 verdict survive unchanged.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94	1.0
896	.091	.155	.208	.032	.013	-.029	.016	.004	.043
(SEM)	.013	.047	.047	.024	.024	.021	.024	.021	.011
768	.149	.238	.324	.206	-.034	-.032	.011	.015	.092
(SEM)	.014	.034	.049	.045	.042	.047	.022	.034	.012
512	.340	.525	.599	.598	.312	.220	.081	.166	.329
(SEM)	.029	.048	.073	.050	.045	.046	.056	.061	.030

power in every band the coarse grid cannot represent — the high- σ persistence of axis 2 in §4.5, which no member of the null family can produce. There is no noise level at which a $2\times$ -downscaled latent yields the native gradient.

Table 3: The floor, debiased. **Endpoint:** at $\sigma=1$ the input is exactly ϵ on every grid, so the input-information term vanishes by construction. **x-zero:** the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\text{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: $N=12$, $D = 4 \dots 64$; x-zero: $N=40$, $D = 4 \dots 32$); raw single-estimate endpoint reads ($D=16$, $N=40$) retained for comparison. The endpoint and x-zero columns agree within CI at every route, and the account’s pre-registered kill switch (all endpoint gaps ≈ 0) does not fire.

route	raw endpoint ($D=16$)	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]	x-zero $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	0.85	1.005 [0.994, 1.016]	—
re-encoding control	−0.038	−0.003 [−0.017, +0.008]	—
1024→896	−0.009	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.127	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.326	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

The debiased map also revises two raw reads, in opposite directions. The 768 route’s high- σ plateau (0.12–0.16 raw) was roughly half estimator bias: debiased means sit within noise of zero across $\sigma \in [0.56, 0.94]$, but with upper bounds that never cross below the 0.02 margin at this (N, D), and a real endpoint floor of $+0.092 \pm 0.012$ remains — so “never safe” softens to *no certifiable window at instrument resolution* (§7.1). Conversely the 896 route’s endpoint, in band raw, shows a small *real* gap ($+0.042 \pm 0.011$) that single-estimate variance had buried; its safe window is the interior $\sigma \in (0.5, 0.94]$, not the endpoint itself.

Ratio does not transfer — the null’s blind spot, measured. Re-running the probe one tier down (896-tier natives, demote to 768/512) fails its pre-registered bar: 896→768 (ratio 0.857, nearly identical to the passing 0.875) retains a high- σ residual of 0.06–0.12 (raw), roughly 2× the 1024→896 plateau and outside the re-encoding band, while the two-tiers-down control stays elevated everywhere (Appendix Table 7). This run predates the debiased instrument, and by the variance model of §3.2 roughly half of that residual is expected to be estimator bias; the separation itself does not rest on it, because the debiased floors of the *same target grids* reproduce it independently — +0.02–0.04 on the 896-token-count grid versus +0.06–0.09 on the 768 grid (§5.2). This is exactly property N2’s blindness made concrete: the null assigns these two routes boundaries within 0.01 of each other at every tolerance, and the measurement separates them into *safe* versus *no certifiable window*. “One tier down at high σ ” is not the invariant; §5.4 locates why (the floor is governed by absolute size, and the 768-grid floor alone fills the 0.02 margin).

5.2 THE FLOOR EXISTS, IS σ -INDEPENDENT, AND IS NOT CARRIED BY THE INPUT

Two designated probes (Table 3) establish the graph term — and with it assumptions A1–A2 of §4.2. At the $\sigma = 1$ *endpoint* the input is pure ϵ — identical in distribution across arms — so the input-information term is zero by construction. The debiased floors are +0.019/+0.056/+0.304 for 896/768/512; the 512 value clears the pre-registered confirmation bar ($\widehat{\text{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually — the token-count floor is not an estimator artifact. The debiasing cuts in both directions across routes: the published raw 768 floor (0.127) was roughly half variance bias, while the raw-in-band 896 endpoint concealed a small real floor.

The *x-zero* probe removes the image from input *and* target everywhere, isolating pure graph-shape sensitivity — and its debiased draw-limit gaps are statistically equal to the endpoint gaps at *every* route (Table 3, last two columns). The endpoint gap *is* the graph floor. This equality is itself a debiasing product: in raw units the 768 endpoint (0.127) sat 2× above its x-zero read (0.064), suggesting a large target-content share (the target $v = \epsilon - x$ still carries x at $\sigma=1$); that share was estimator bias. A target-strength sweep at the endpoint (target $\epsilon - \alpha x$, $\alpha \in [0, 1]$) is the designated cheap confirmation, with predicted slope ≈ 0 [pending].

How does the floor relate to the interior high- σ bins? For 512 the debiased window means (+0.08–0.60) sit at or above the floor, as additivity requires. For 896 and 768 the floors (0.02–0.04 and 0.06–0.09) are *at or below* the interior bins’ detectability threshold ($\epsilon^* = 0.03$ –0.08 at $N=40$, $D=8$): the endpoint sweeps resolve them only because their effective draw budget is an order of magnitude

larger. The raw claim “the high- σ plateau is the floor” therefore sharpens to: the floor is what the plateau converges to when the input term dies and the estimator is given enough draws to see it.

Two further exonerations: the re-encoding control sits at zero within CI — debiased -0.003 $[-0.017, +0.008]$ — so the VAE round trip is not the cause (and, the converse asymmetry: the round trip has *measurable* input-level error, yet its gradient cost is below instrument resolution — input-level error neither implies nor excuses gradient-level cost); and a forward-only probe of the model’s caption-conditioned *prior* across grids shows route distances that are flat where the floors are strongly ordered — so the floor’s route-ordering lives in the Jacobian factor J , not in the model’s resolution-conditioned prior $\hat{x}_{\text{prior}}$, and not in a training-distribution discontinuity (the never-trained 1280 grid is no more distant from 1024 than heavily-trained adjacent pairs are from each other).

The floor as a ledger. Combining the designated probes, the floor decomposes additively, each share measured by its own intervention:

$$\text{Floor}_e = \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\approx 0 \text{ by x-zero equality}} + \underbrace{\text{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\text{Resid}_e}_{\text{remainder}}$$

with debiased totals 0.02–0.04 (896; below the PI probe’s paired resolution, undecomposed), ≈ 0.07 (768; the paired PI erasure of $+0.081$ — a same-grid contrast that cancels draw bias — accounts for essentially all of it, $\text{Resid} \approx 0$), and ≈ 0.30 (512; $\text{RoPE} \approx 0.10$, $\text{Resid} \approx 0.20$) — the positional decomposition itself is §5.5. One honest caveat on the first share: the re-encoding control (native decode→re-encode) is a *proxy* for the pipeline cost demotion actually pays (source downscale→encode); a demote→re-promote arm would close that gap empirically and is left as designed future work.

5.3 THE INPUT BRANCH: A ROUTE-UNIFORM NUMERATOR OVER A U-SHAPED DENOMINATOR

$m(\sigma)$ is universal — the residual-level null fails. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_\epsilon [\hat{y} - (\epsilon - x)]$ per grid and σ — the exact r in $g = J^T r$ — shows a strong, monotone σ -shape (cross-grid excess $0.89 \rightarrow 0.36$ in relative L_2 from $\sigma=0.125$ to 1) that is the *same for every route* within ± 0.02 , across routes whose gradient floors span 0 to 0.3 and whose crossovers span 0.5 to nonexistent (Appendix Table 8). This kills the diagonal null at the residual level (N4): with no cross-frequency coupling, its mismatch is confined to the destroyed band and ordered by destroyed-band energy, which differs substantially across these routes — route-uniformity says the real mismatch is carried by cross-band structure the diagonal model cannot express. All route identity therefore lives in J , not in the prediction error — a pre-registered null that would equally have falsified the two-term account had the residual curves been route-ordered. The decay is smooth, Wiener-like posterior shrinkage with no hard gate at the spectral crossover: gap curves read as “collapsed” where S_1 sinks below the instrument’s detectability threshold, not at σ_{eq} .

$G(\sigma)$ shapes the curve. Because the gap is a cosine it measures a mismatch *fraction*; the total gradient norm is U-shaped in σ ($2.6 \rightarrow 0.4$ at $\sigma \approx 0.3 \rightarrow 9.3$; Table 1, last row; Fig. 1d) — large at high σ where the model is pulling composition out of the prior and the residual is image-scale, small in the mid- σ refinement regime, and inflated again at low σ by irreducible ϵ . A monotone absolute mismatch divided by a U-shaped norm yields exactly the measured interior-peaked gap curves: bin-mean gap tracks $1/\|g\|$ (Spearman $+0.55$ to $+0.90$ across arms and runs, and within images), and renormalizing by G flips the anomalous low- σ dip into the monotone maximum in every arm of both runs — axis 5 of §4.5, resolved for the two-term account. This is a post-hoc consistency analysis over the raw curves (flagged as such): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

5.4 THE GOVERNORS: RATIO SETS THE AMPLITUDE, ABSOLUTE SIZE SETS THE FLOOR

With the floor isolated, what orders the routes? Two probes give a clean division of labor — axis 3 of §4.5. The 1280-tier runs predate the debiased instrument, but the finite-draw bias is strictly *positive* (§3.2): an in-band or ≈ 0 raw read is therefore conservative — bias could only have inflated it — while raw positive values carry some inflation that does not affect the ordering reads below.

Table 4: Iso-severity probe (**raw estimator**): gap by σ -bin on the 1280-tier cache ($N=24$, 4 draws/bin) against the corpus 1024 $\rightarrow$ 896 curve. The ratio-matched routes coincide within ~ 1.4 combined SEM at every bin despite a $1.6\times$ difference in absolute target size; the same-native harsher-ratio route separates. *896’s canonical 16-draw endpoint is -0.009 raw, $+0.019$ debiased (Table 3).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280 $\rightarrow$ 1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280 $\rightarrow$ 1024 (ratio .80)	.170	.178	.111	.002	.061
1024 $\rightarrow$ 896 (ratio .875)	.132	.076	.077	.049	.100*

Absolute size, not ratio, sets the floor. The route 1280 $\rightarrow$ 1024 is *more* aggressive by ratio (0.80) than 896 $\rightarrow$ 768 (0.857, no certifiable window), yet its floor is ≈ 0 (a bias-conservative read) and it enters the re-encoding band at $\sigma \gtrsim 0.75$ — refuting any pure ratio rule for the floor, and with it the null’s only degree of freedom. Conversely the floor grows monotonically as the *target* grid shrinks (debiased 0.02–0.04 / 0.06–0.09 / ≈ 0.30 at target $\sim 3,000/2,160/1,000$ tokens), consistent with “floor = how well the coarse graph approximates the fine graph’s computation.”

Ratio, not absolute size, sets the amplitude A_e . A synthetic iso-severity route 1280 $\rightarrow$ 1120 — edge ratio 0.875 exactly matched to 1024 $\rightarrow$ 896, but $1.6\times$ its absolute target size — reproduces the 1024 $\rightarrow$ 896 curve bin-for-bin and its crossover $\sigma^* \approx 0.5$, while the same-native 1280 $\rightarrow$ 1024 (ratio 0.80) separates and floors later (Table 4). Amplitude follows ratio; with $\text{Floor}_{1120} \approx 0$ this completes the division: *ratio sets A_e ; absolute target capacity sets Floor_e* . The pre-registered contrast (ratio-governed $\Rightarrow \sigma^* \approx 0.5$; capacity-governed $\Rightarrow$ earlier floor) discriminated cleanly. Note what survives of the spectral account here: ratio *is* the right governor for the input branch — the null’s error is not its governor but its scope, mistaking one term for the whole gap.

5.5 WHERE THE FLOOR LIVES: DEPTH, AND A POSITIONAL DECOMPOSITION

Depth, not module type. Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) shows the floor concentrated in *depth*: early blocks (0–9, peaking at 3–8) carry $\sim 3\times$ the late-block gap, uniformly across every module type within a block (at 512/ $\sigma=1$, every type sits in 0.22–0.31; Appendix Table 9). The content share (endpoint minus x-zero) is a late-block minority effect. The mechanism picture: the first ~ 10 blocks build grid-calibrated token statistics; the divergence propagates into every parameter type’s gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Notably, query and key projections show *zero* excess over value projections — which we initially read as refuting a rotary-embedding mechanism. That inference was wrong, in an instructive way.

$\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$. Landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. The discriminating experiment is the origin-side intervention of §4.4: evaluate the demoted grid with positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\text{nat}}/H_{\text{dem}}$ per axis), making its relative phase geometry exactly match native. At the $\sigma=1$ endpoint this *erases* the 768 floor ($+0.080 \rightarrow -0.001$ raw; paired $\Delta = +0.081$, 78% of images improved) and removes $\sim 30\%$ of the 512 floor ($+0.320 \rightarrow +0.224$ raw), while leaving the in-band 896 control untouched (Appendix Table 10). The paired Δ s are same-grid contrasts at matched draws, so the finite-draw bias cancels to first order and they survive re-anchoring in debiased units: against the debiased 768 floor of $+0.056$ – 0.092 , the $+0.081$ erasure accounts for essentially all of it ($\text{Resid}_{768} \approx 0$); against the debiased 512 floor of ≈ 0.30 , the $+0.096$ erasure leaves a $\text{Resid}_{512} \approx 0.2$ bulk. The mild-route floor is mostly a *coordinate-system artifact*; the harsh-route bulk is a genuine non-positional graph residue (attention-over- N statistics, sequence-length-dependent normalization, capacity). The capacity governor of §5.4 attaches to Resid_e , as the band-counting sketch anticipated. The 896 route’s small debiased floor (0.02–0.04) sits below the PI probe’s paired resolution and remains undecomposed.

The positional share is a noise-regime artifact only. A σ -resolved follow-up closes the tempting route this opens (training 1024 $\rightarrow$ 768 through interpolated coordinates): with image content in the input, the stretched forward is *off-manifold* — the interpolated arm is *worse* than the plain de-

Table 5: The spectral null (Eq. 7) instantiated on the measured latent spectrum: predicted safe boundaries t^* per route as the tolerance δ sweeps its range. SPD’s published default is $\delta = 0.01$; the equal-power slice is the classical crossover. No row reproduces the measured pattern (last line; §7.1): the δ that matches the safe route’s boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.05	0.41	0.43	0.53	
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	SPD’s published default
0.005	0.69	0.71	0.79	
measured, debiased	≈ 0.5	none certified	unsafe, all σ	floors .02–.04 / .06–.09 / $\approx .3$

moted arm through $\sigma \in [0.56, 0.81]$ (paired -0.05 to -0.11) and wins only at $\sigma \geq 0.94$. The floor’s positional share is erasable only where the input is noise-dominated, so the decomposition is a mechanism finding, not a training lever — and the input-branch term of the 768 route is fatal on its own ($+0.09$ – 0.22 over control across the window). A frequency-*selective* banded alignment does survive in-window (§7.2).

6 THE NULL READ IN GAP UNITS

The confrontation is stated last because it consumes the measurements: the null’s tolerances are instantiated on the measured spectrum (§5.1), and reading them in gap units requires the measured $G(\sigma)$ and a route gain calibrated on the safe route.

6.1 THE TOLERANCE FAMILY ON THE MEASURED SPECTRUM

Table 5 instantiates Eq. 7 on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065). Because the latents are high-frequency-quiet, P barely varies across the cuts and the family is *compressed*: the spread between the mildest and harshest route never exceeds ~ 0.13 in σ at any tolerance.

Which tolerance? Eq. 7 needs a δ , and it is worth asking where one could legitimately come from. SPD’s published $\delta = 0.01$ is an inference-side choice, selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026) — nothing ties it to gradients. Machine precision is not the relevant floor either: bf16 rounding on unit-scale per-band amplitudes injects power of order 10^{-5} or below, orders of magnitude beneath any boundary-moving tolerance. Our setting, however, fixes a tolerance *by construction*: the demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$ — the same pipeline noise the re-encoding control prices at the gradient level — and D is expressed in exactly Eq. 7’s units (per-band power against unit noise). Band content below $D(f_{\text{cut}})$ is invisible to the substitution pipeline even with no resolution change, so $\delta_{\text{reenc}}(e) := D(f_{\text{cut}}(e))$ anchors the null with zero free parameters. On the probe set this floor and its implied boundaries are **[pending: reenc_noise_floor.py, queued]**; we will report the anchored row alongside the family in Table 5. Either way the confrontation does not hinge on the choice: the structural failures (N2, N3) are δ -independent.

6.2 SCORING THE ACCOUNTS

Against the five axes of §4.5, the null loses each one: the boundary pattern is $\{\approx 0.5, \text{no certifiable window, certifiably unsafe everywhere}\}$ (axis 1, §5.1); gaps persist at pure-noise input, equal to the floor and to the image-free control (axis 2, §5.2); ratio governs the amplitude while absolute size governs the floor (axis 3, §5.4); the residual curves are route-uniform under route-ordered gaps (axis 4, §5.3); and the curve shape follows m/G^p , not per-band SNR (axis 5, §5.3). Taking Eq. 6 with measured m , G , A_e , Floor_e , the account also retrodicts the observations that motivated no probe: (i) $\sigma^* \approx 0.5$ rather than 0.14 — the crossover is where $A_e m/G^p$ sinks below the instrument’s

Table 6: The measured safety map — *per-example* object (the worst case, what a batch-1 user sees). Floor_e = debiased endpoint floor where the debiased instrument ran; † marks raw-instrument-only routes (the finite-draw bias is strictly positive, so their ≈ 0 floors and in-band verdicts are conservative reads). σ^* = noise level above which the paired debiased gap is non-inferior at instrument resolution (§3.3); token ratio = per-forward cost of the demoted grid.

route	edge ratio	target tokens	Floor_e	σ^*	verdict
1024→896	0.875	~3,000	0.02–0.04	≈ 0.5	safe , $\sigma \in (0.5, 0.94]$
1280→1024 †	0.80	~4,100	≈ 0	$\in (0.625, 0.875)$	safe , $\sigma \gtrsim 0.75$
1280→1120 †	0.875	~4,800	≈ 0	≈ 0.5	safe (synthetic grid)
896→768 †	0.857	~2,160	0.06–0.12 raw	—	no certifiable window
1024→768	0.75	~2,160	0.06–0.09	—	no certifiable window
any →512	≤ 0.57	~1,000	≈ 0.30	—	certifiably unsafe, all σ

detectability threshold, a property of the ratio, so the $3.5\times$ discrepancy is expected, not anomalous; (ii) the 896→768 failure (the 768-grid floor alone fills the certifiable margin — no σ -gate can rescue it); (iii) 1280→1024 flooring *later* ($\sigma^* \approx 0.75$) but cleanly (harsher ratio $\Rightarrow$ larger A_e ; large target $\Rightarrow \text{Floor} \approx 0$) — the single observation that breaks both pure-ratio and pure-capacity route-ordering stories.

A corollary worth stating plainly: the safety of 1024→896 is an *empirical smoothness statement about this network’s function across nearby token counts*, not an information-theoretic statement about the data. There was never a reason to expect a universal ratio invariant.

6.3 FROM TOLERANCES TO PREDICTED GAP CURVES

Boundary comparison undersells the confrontation: a tolerance δ determines not just where the null’s gap vanishes but a full predicted *curve*. Under the diagonal model on the measured $P(f)$, each δ implies a destroyed-band Bayes-residual mismatch $m_{\text{null},e}(\sigma; \delta)$; mapped through the measured $G(\sigma)^P$ with the route gain A calibrated on the one safe route, it becomes a predicted gap curve overlayable on Table 2’s measured ones. Unit honesty: the null emits residual units only — the G^P bridge and the calibrated A belong to the two-term account, so this is *the null read through our bridge*, and §4 is load-bearing for the confrontation rather than decorative. The overlay figure and the continuous $t^*(\delta)$ sweep (three near-coincident route curves, family spread ≤ 0.13 , measured boundaries as markers, δ_{reenc} as the anchor line) are **[pending: analysis over existing rows; the anchored line additionally waits on reenc_noise_floor.py]**.

7 FROM THE GAP TO LOW-RESOLUTION TRAINING

7.1 THE (ROUTE, σ) SAFETY MAP

Table 6 is the practical summary of §5: what replaces the spectral rule is not a corrected formula but a measured map at a stated resolution. Safety is a per-route boundary σ^* (route), jointly determined by ratio (through A_e) and absolute target size (through Floor_e); neither alone predicts the ordering. For the corpus safe route the fine print is: every bin in $(0.5, 0.94]$ has a debiased mean within ± 0.03 of zero, formal certification at the strict 0.02 margin is achieved at $\sigma = 0.688$ (elsewhere bin-level ε^* exceeds it at the current N, D), and the $\sigma=1$ endpoint itself carries a small real gap ($+0.042 \pm 0.011$) — the safe window is the interior.

The map depends on the aggregation object. “Safe for whom” matters: the per-example map above is the worst case, and the object SGD actually follows — per-image gradients summed over the batch before the cosine — is more forgiving. Pooled-arm probes (pool size 4) collapse the 1024→896 gap to ≈ 0 at every bin $\sigma \geq 0.625$ and the 1024→768 gap to ≈ 0 at $\sigma \geq 0.875$: the per-example and batch-aggregate objects genuinely disagree at high σ , so any safety claim must state which object it is a claim about. The small per-image residual behaves as draw noise that cancels across images rather than as a shared bias. We publish the per-example map as the verdict (it bounds every batch size from below; pooled raw reads are bias-conservative in the in-band direction but pooled arms do not yet have

self-floors), and a debiased verdict grid at the shipped trainer’s actual batch $\times$ accumulation operating point is designed [pending]. A redundancy-stratified analysis finds no per-image demotion-targeting lever (null trend across strata), consistent with the floor being an image-generic network property.

7.2 σ -CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured — no extrapolation beyond the map:

σ -first draw. The trainer draws each batch’s σ *before* fetching latents, from the unchanged training density (the σ marginal is untouched — demotion conditions on σ , never the reverse). When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe’s own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream — noise, target, masking, loss — derives from the swapped latent. Off-route images always train native, as does validation. The small real gap at the exact $\sigma=1$ endpoint (§5.2) does not bite here: it is a boundary point of measure zero for the continuous training σ -density, and the last interior bin ($\sigma \in (0.875, 1)$) reads ≈ 0 debiased.

Banded rotary alignment, σ -gated. Motivated by §5.5, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, ramp between, with the band thresholds scheduled by a sigmoid gate in σ following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs — it erases the static variant’s low- σ liability ($+0.064 \rightarrow +0.033 \pm 0.025$) while preserving its high- σ gains (best and most stable arm at $\sigma \geq 0.59$ and the endpoint, paired -0.05 vs plain demotion). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$.

Cost accounting. With $\sim 50\%$ of draws above the gate and the demoted forward at $\sim 0.72\times$ tokens, the fixed-step epoch cost is $\approx 0.5 \cdot 0.72 + 0.5 = 0.86$ — a *projected ceiling* of $\sim 14\%$ wall-clock on our corpus (96% of records are on the safe route), scaling with high-tier content; the realized saving (including cache-load overheads) is measured only by the pending A/B below. The pitch is wall-clock at fixed steps, never “more steps in the same time.”

Weight-space sanity. Under bit-exact deterministic training (paired runs; without determinism, twin runs put a chaos floor of 0.41 on checkpoint-displacement cosine, which would swamp the read), the rotary refinement adds no displacement over the demotion arm it refines (base $\leftrightarrow$ refined 0.319 vs base $\leftrightarrow$ demoted 0.320), with its footprint concentrated in the low-signal early/mid blocks — consistent with §5.5’s depth profile.

What this section does not claim. The end-to-end acceptance gate — a fixed-step A/B at training scale scoring generation-quality non-inferiority (CMMD; Jayasumana et al., 2024) against the realized wall-clock saving — is designed and pre-committed (a regression closes the line) but not yet run; all claims above are gradient-level and weight-space-level. We flag this deliberately: the gap between gradient-level equivalence at $N=40$ and thousands of optimizer steps is exactly the kind of gap this paper is about.

8 LIMITATIONS

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint first is the designated bound on how far the map generalizes. The floor’s dependence on the fine-tuning operating point through J is untested (the designated probe: fine-tune at the 1280 tier, re-probe 1280 $\rightarrow$ 1024). Relatedly, the probe pool’s relationship to the adapter’s own fine-tune set was never a controlled axis (2 of 40 verdict stems are in it); a controlled-LoRA probe with a frozen membership manifest (trained-on / held-out / near-OOD cells) is designed to measure this dependence rather than concede it [pending].

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed only by the pending A/B above.

The account is a form, not a fit. Eq. 6’s value is that each object is independently measurable and every divergent prediction discriminated cleanly; we fit no coefficients, and the confrontation with the residual-level null (N4) is structural (ordering and floor), not a quantitative fit of its σ -shape. Likewise the band-counting sketch for RoPE_c/Resid_c is calibrated, not derived.

Instrument resolution. “Safe” is one-sided non-inferiority at the measured detectability threshold (§3.3): $\varepsilon^* = 0.03\text{--}0.08$ per bin at the verdict grid’s (N, D) , $\approx 0.01\text{--}0.02$ at the endpoint sweeps. True gaps below ε^* are undetectable by construction, the σ^* boundaries are localized only to a bin, and every “no certifiable window” claim is (N, D) -relative — a larger campaign could certify (or refute) the 768 high- σ window whose debiased means already sit near zero.

Debiased coverage. The debiasing campaign covered the corpus verdict grid: the endpoint draw sweeps, the 8-bin map, and x-zero. The 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention runs remain raw-instrument reads, defended where used by the one-signed-bias argument (a raw ≈ 0 read is conservative) and by paired same-grid contrasts (the bias cancels to first order); pooled arms have no self-floors yet.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ -gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

9 CONCLUSION

“Almost equivalent” input signal is not almost-equivalent training signal — and measuring the difference honestly is itself a metrology problem. Spectral sufficiency of the noisy input does not guarantee gradient equivalence under resolution substitution: to first order the training-side cost of demotion has two terms, and the spectral argument — even in its strongest tolerance-parameterized form — is a model of only one of them, a one-parameter, ratio-only family that cannot express the σ -independent floor the network’s grid-calibrated computation carries. The debiased measurements bear the structure out at every point of divergence: the mildest route’s crossover sits at $3.5\times$ the equal-power prediction and no tolerance reproduces the map; a $2\times$ -downscaled latent is certifiably unsafe at every noise level — because expected gradients are not noise-masked and the network function itself is grid-calibrated in its early blocks, partly through its positional coordinate system and partly through an irreducible dependence of the computation on token count; and the surviving endpoint gap equals its image-free control everywhere, placing the whole floor in the compute graph. The metrology cut both ways: naive finite-draw gap estimation manufactures exactly the token-count-ordered floor signature under test, half the intermediate route’s raw floor was estimator bias, and our own headline claims were re-derived — and in one case softened — under the correction. What replaces the spectral rule is a measured map at a stated resolution: per-route boundaries governed by downscale ratio on the noise-dependent term and absolute grid size on the noise-independent one. On the one route our corpus makes safe, the map projects a ceiling of $\sim 14\%$ wall-clock at fixed steps, pending the end-to-end A/B. We offer the account and its instrument — per- σ -bin gradient agreement with a redraw floor, a re-encoding control, per-arm self-floors, and a stated detectability threshold, criteria frozen before data — as the cheap, general test that any future “train part of the time at lower resolution” proposal should pass before it is believed; without the debiasing, that test manufactures the very effect it looks for.

REPRODUCIBILITY STATEMENT

The probe scripts (`run_sigma_probe.py` — including the `--self_floor` and `--draw_sweep` debiasing modes — `run_prior_distance.py`, `rapd.py`, `compare_ckpt_dw.py`), the trainer wiring (`--sigma_lowres`), and the frozen pre-registration documents (including the debiasing campaign’s decision rules) are in the public repository at https://github.com/sorryhyun/anima_lora (project line `project/sigma_lowres/`). The debiased verdict runs’ complete per-image records (`result.json` + `per_image.jsonl`) ship in-repo under `project/sigma_lowres/paper_bench/runs/`; the earlier raw-instrument run archives are excluded by the repository’s results ignore rule and will be published as a companion tarball [pending] — until then, raw-run numbers are reproducible from the scripts but their original records are not themselves public. Paired training runs use a `--deterministic` mode (deterministic attention back-

ward included) verified bit-identical across twin runs; finite-draw cosines are kernel-path-sensitive, so all cosine pairings are computed within a single run (Appendix A).

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A INSTRUMENT DETAILS

Estimator. Per image, arm, and σ -bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at $D=8$ are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is never used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in raw runs its gap served as the validity band (± 0.04 ; observed $|\cdot| \leq 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §3.2. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer’s σ -density reproduces earlier σ -marginalized measurements from the same instrument family (consistency check).

Debiasing (§3.2), implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 2 is then computed per image and bin, and the map read is the paired difference $\Delta_i = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\Delta_i| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \leq 32$ sets; prefix sums, no extra forwards) and $\text{gap}(D) = \text{gap}_\infty + c/D$ is fit on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path chaotic: twin runs sharing a warm compiler kernel cache agree to $|\Delta \cos| \leq 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at $D=2$ — so gap/floor/self-floor pairings are never compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

Endpoint and x-zero modes. The $\sigma=1$ endpoint bin feeds input exactly ϵ (the input-information term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low- σ bins are off-manifold and the $\sigma=1$ read is primary. In that regime the residual is $\approx -\hat{x}_{\text{prior}}$, so “graph-dominated” includes the model’s grid-conditioned prior; the prior-distance probe (§5.2) subsequently dissociated the prior from the floor’s route ordering.

Spectral-null computation. Table 5 evaluates Eq. 7 at the demoted grids’ Nyquist frequencies $f_{\text{cut}} = \frac{1}{2} e/e_{\text{nat}}$ on the $N=40$ mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065). The equal-power slice $\delta = (1 + P_{\omega})/2$ recovers $\sigma_{\text{eq}} = \sqrt{P}/(1 + \sqrt{P})$ exactly. Across $\delta \in [0.001, 0.5]$ the mildest–harshes spread never exceeds 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) never separate by more than 0.01. The parameter-free anchor δ_{reenc} of §4.3 is measured by the same estimator: RAPSD of the re-encode arm’s latent error (fresh resize–encode minus cached latent, the gradient probe’s own control chain) on the same probe set, interpolated at each route’s cut frequency [**pending: reenc_noise_floor.py**].

B ADDITIONAL TABLES

Table 7: Ratio-transfer run (**raw estimator**): 896-tier natives ($N=40$), pre-registered bar “gap₇₆₈ within the re-encoding band at $\sigma \geq 0.5$ ” — **fails** (residual 0.06–0.12, $\sim 2\times$ the 1024→896 plateau; $\sim$ half of it is expected estimator bias, §3.2 — the separation is independently confirmed by the debiased same-grid floors, §5.2).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 8: Route-uniformity of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_{\epsilon} [\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2 ; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 9: Depth localization of the floor (**raw estimator**): mean per-block gap at $\sigma=1$, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry $\sim 3\times$ the late-block gap. The apparent content share (ep – xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adaptor level (Table 3).

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 10: Positional-interpolation intervention at the $\sigma=1$ endpoint ($N=40$, 16 draws; **raw** arm values — the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and $\sim 30\%$ of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024 $\rightarrow$ 896 (control)	-0.021 (.041)	-0.040 (.040)	—	—
1024 $\rightarrow$ 768	+0.080 (.048)	-0.001 (.039)	+0.081 (.031)	78%
1024 $\rightarrow$ 512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%